A FLAT MORPHISM LOCALLY OF FINITE PRESENTATION OF ALGEBRAIC SPACES IS UNIVERSALLY OPEN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is one sentence with no numbered parts; the proof is the standard presentation-and-transfer argument and is complete.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Lifting a morphism of algebraic spaces to a diagram of schemes. Spaces, Lemma spaces-lemma-lift-morphism-presentations, used at proof lines 6102-6111. Produces U, V and the morphism alpha.
- Flatness and local finite presentation are local on the source and target. Lemmas lemma-flat-local and lemma-finite-presentation-local, used at proof lines 6112-6114. Make alpha flat and locally of finite presentation.
- The scheme case: flat plus locally of finite presentation is universally open. Morphisms, Lemma morphisms-lemma-fppf-open, used at proof lines 6115-6117. Gives universal openness of alpha.
- The local criterion for universal openness of a morphism of algebraic spaces. Lemma lemma-universally-open-local, used at proof lines 6118-6119. Transfers universal openness from alpha to f.

Source reference. Original lemma 0410: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5275–5275. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03MM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4754–4754. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03Z5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5208–5208. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01UA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2403–2403. Navigation pointer into the Stacks

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Source reference. Original lemma 02X1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3824–3824. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *A flat morphism locally of finite presentation is universally open.*

Proof. Let $f : X \to Y$ be a flat morphism locally of finite presentation of algebraic spaces over S. Choose a diagram

$$\begin{array}{ccc} U & \xrightarrow[\alpha]{} & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

where U and V are schemes and the vertical arrows are surjective and étale, see Spaces, Lemma 02X1. By Lemmas 03MM and 0410 the morphism α is flat and locally of finite presentation. Hence by Morphisms, Lemma 01UA we see that α is universally open. Hence $X \to Y$ is universally open according to Lemma 03Z5. $\square$